

CSI ORIGIN 2026

PROBLEM STATEMENT

1 PROBLEM STATEMENT

Hackathon / Event Name : CSI ORIGIN 2026

Problem Statement ID : 7

Title : Securing AI Agents That Can Independently Control Financial Assets

2 BACKGROUND

Current Situation / Context

As AI agents evolve from conversational assistants into autonomous economic actors, they are increasingly capable of accessing financial accounts, holding digital wallets, initiating transactions, allocating capital, interacting with financial protocols, and negotiating with other agents.

Traditional financial security models generally assume that a human or trusted application initiates and authorises financial transactions. Autonomous financial agents challenge this assumption because the agent itself becomes an active financial actor with permissions, decision-making capabilities, and potentially direct access to financial assets.

Underlying Problem or Need

The core gap is the absence of a security and control layer specifically designed to govern autonomous agents that can independently affect financial assets.

A financial agent may be legitimately authorised to perform certain actions, yet an individual action may still be unsafe, manipulated, anomalous, or inconsistent with the agent's expected behaviour. Security therefore needs to evaluate not only whether an action is technically valid, but whether it is consistent with the agent's identity, authority, intent, policies, context, and historical behaviour.

Why This Is Becoming Important

As autonomous agents gain greater authority over financial accounts, wallets, capital allocation, and financial protocols, the consequences of agent compromise or manipulation become significantly more severe.

A compromised or manipulated agent could misallocate capital, execute unauthorised transactions, fall victim to prompt or instruction injection, interact with malicious contracts, transfer assets to fraudulent counterparties, or make financially harmful decisions while technically operating within its granted credentials.

Limitations of the Current Situation

Conventional transaction security can establish whether a transaction is properly authenticated, correctly signed, or technically valid. However, these checks do not necessarily establish whether the transaction represents legitimate or expected agent behaviour.

Static permission systems may also provide excessive authority to autonomous agents or require human approval for every action, which undermines the benefits of autonomy.

There is therefore a gap between transaction validity and behavioural legitimacy.

Impact of Leaving This Unresolved

Without an appropriate security and control layer, autonomous financial agents may become high-impact attack surfaces with direct access to financial assets.

A compromised agent could potentially execute legitimate-looking but harmful transactions, resulting in financial loss, unauthorised transfers, policy violations, or interaction with malicious counterparties and protocols.

Conversely, overly restrictive security mechanisms could make autonomous financial agents impractical by requiring continuous human intervention.

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What Is Going Wrong / Missing?

Existing financial security mechanisms are not designed to continuously determine whether an autonomous agent's intended financial action is safe and consistent with its authorised role and expected behaviour.

A transaction can be correctly signed and technically valid while still being inconsistent with the agent's normal behaviour, spending policies, authorised counterparties, risk limits, or intended purpose.

The missing capability is a security layer that can interpret agent intent, evaluate contextual risk, detect behavioural deviations, enforce policies, and determine whether an action should be autonomously executed, constrained, escalated, or blocked.

Who or What Is Affected?

The problem affects autonomous AI agents that control financial accounts, wallets, investment capital, payment systems, financial protocols, or other financial assets.

It also affects organisations and financial systems that grant autonomous agents authority to execute transactions or make financial decisions on their behalf.

Why Is It Difficult or Important?

The system must secure an agent without eliminating its autonomy.

Requiring human approval for every transaction may provide strong control but defeats the purpose of autonomous execution. Conversely, granting unrestricted authority exposes financial assets to potentially severe consequences if the agent is compromised, manipulated, or behaves unexpectedly.

The system must therefore determine appropriate intervention based on contextual risk. Low-risk actions may be executed autonomously, while increasingly risky or anomalous actions may require additional verification, constrained execution, or human escalation.

Consequences

If this problem remains unresolved, autonomous financial agents may either operate with excessive financial authority or require excessive human supervision.

The former increases the potential impact of compromised or manipulated agents, while the latter limits scalability and prevents agents from achieving meaningful financial autonomy.

4 CHALLENGE

Design and build a security and control layer for autonomous financial agents that continuously evaluates whether an agent's intended action is authorised, contextually appropriate, and sufficiently safe before allowing it to affect financial assets.

The system should interpret agent intent, evaluate contextual and behavioural risk, detect threats or deviations, enforce financial policies, and apply graduated responses ranging from autonomous execution to constrained execution, escalation, or blocking.

The System Should Operate as a Continuous Security Loop

Agent Intent → Interpret Action → Verify Identity → Evaluate Context → Detect Threats → Enforce Policy → Approve / Reject / Escalate → Monitor Outcome

Problem Requirements / Constraints

1. The system must evaluate financial actions based on agent identity, authorisation, transaction intent, permissions, and applicable financial policies.
2. The system must consider contextual risk factors including transaction amount and frequency, counterparty characteristics, transaction history, smart-contract or protocol risk, and relevant market or liquidity conditions.
3. The system must detect behavioural deviations from an agent's established or expected patterns and distinguish potentially anomalous actions from routine activity.
4. The system must account for malicious or manipulated instructions, including prompt or instruction manipulation that may cause an agent to attempt an otherwise unusual financial action.
5. The system must enforce explicit spending, transaction, exposure, and permission constraints before allowing an action to affect financial assets.
6. The system must support graduated intervention based on risk, including autonomous execution for low-risk actions, additional verification or constrained execution for medium-risk actions, and blocking or escalation for high-risk actions.
7. The system must continuously monitor outcomes and relevant agent behaviour so that security decisions can adapt when the agent's behaviour or surrounding risk conditions change.

5 ANNEXURE — DETAILED RULES / CONSTRAINTS

Agent Identity and Authorisation

The system must establish whether the agent is authorised to perform the intended action and whether the action falls within the permissions and financial responsibilities assigned to that agent.

Transaction Intent

The security layer must evaluate the intended purpose and characteristics of an action rather than relying solely on cryptographic validity or successful authentication.

A technically valid transaction must not automatically be treated as legitimate agent behaviour.

Behavioural Anomaly Detection

The system should establish relevant behavioural patterns for an agent and identify material deviations, including unusual transaction amounts, frequencies, counterparties, destinations, or financial activities.

Counterparty and Protocol Risk

The system must account for relevant risks associated with transaction counterparties, wallets, smart contracts, financial protocols, or other external entities with which the agent interacts.

Prompt and Instruction Manipulation

The system must consider the possibility that an agent's intended action has been influenced by malicious, unauthorised, or manipulated instructions. The security layer should evaluate whether the resulting financial action remains consistent with the agent's authorised objectives and policies.

Policy Enforcement

The system must enforce predefined financial controls such as spending limits, exposure limits, permitted counterparties, transaction frequency constraints, and other applicable policies.

Graduated Autonomy

The system must avoid treating all financial actions identically. Security responses should reflect assessed risk:

Low Risk → Autonomous Execution

Medium Risk → Additional Verification / Constrained Execution

High Risk → Block / Human Escalation

Outcome Monitoring

Security evaluation must continue after an action is approved or executed. Material changes in agent behaviour, transaction outcomes, or external risk conditions should be capable of influencing subsequent security decisions.

6 FINAL PROBLEM CONSTRAINTS

- The system must evaluate whether an action is consistent with an agent's authorisation, intent, expected behaviour, and financial policies, not merely whether the transaction is technically valid.
- The system must account for behavioural anomalies, transaction patterns, counterparties, protocol risks, instruction manipulation, and relevant financial context.
- The system must enforce explicit spending, exposure, permission, and transaction constraints.
- The system must provide graduated autonomy, allowing low-risk actions to proceed autonomously while subjecting higher-risk actions to additional controls, escalation, or blocking.
- The system must continuously monitor agent behaviour and financial outcomes rather than relying exclusively on one-time security checks.
- The security layer must balance financial protection with agent autonomy and must not require human approval for every routine transaction.
- The challenge is focused on enabling risk-aware autonomy for financial agents, not merely building conventional transaction authentication, fraud detection, or wallet security.